FAIR 3D HERITAGE
Documentation, Publication and Preservation of the Digital 3D Heritage on the Web

Lightweight Metadata for 3D Representations of Immovable Cultural Assets and Their Damages: Bridging the Gap Between Data Quality Standards and Participative Documentation Methods

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Abstract

This research is motivated by an urgent need to systematically document Ukrainian cultural heritage in light of the ongoing Russo-Ukrainian war. Preparatory measures, such as inventories of notable assets, were not fully implemented before the war began (Campfens et al., 2023, pp. 65-66). The inaccessibility of up-to-date architectural documentation hinders restoration efforts.

The proliferation of digital artefacts in cultural heritage presents both opportunities and poses additional challenges. Diverse data, ranging from textual descriptions and photographic documentation to architectural drawings and 3D models, are being produced by a wide variety of professionals involved in safeguarding heritage. Even within 3D representations, variations exist in quality, level of detail, and semantic depth: models may range from partial, semantically limited point clouds to richly annotated Industry Foundation Classes (IFC) files.

In addition, several trends are reshaping the domain: Linked Data enables machine-readable representation of complex relationships between events, actors, sites, and their digital counterparts; Historic Building Information Modelling (HBIM) is maturing into a conservation methodology (Penjor et al., 2024); widely available and constantly improving consumer technologies facilitate high-quality data acquisition, including 3D reality capture. Moreover, in heritage preservation, a shift in paradigm is taking place, characterised by the growing prominence of community-oriented approaches (Campfens et al., 2023, p. 17).

In the Ukrainian context, crowdsourcing represents a promising strategy for large-scale (remote) data collection, as it helps mitigate the shortage of specialists and financial resources that frequently accompany armed conflicts. However, the resulting multiplicity and variety of contributors and data require approaches that ensure traceability of evidence and compliance with FAIR (Findable, Accessible, Interoperable, Reusable) principles (Wilkinson et al., 2016).

In large-scale and time-sensitive contexts, scalable solutions are needed for both distributed data acquisition as well as its efficient search, retrieval, and reuse. This calls for the adoption of machine-readable structures and open formats for coordinated documentation. Rich metadata allows professionals to pre-evaluate suitability of 3D representations without inspecting each file individually and select data tailored to specific analytical needs from a vast pool of documents, thus, reducing manual effort. For example, architects may look for detailed as-planned models with an information about structure´s materials, while a law enforcer would rather be interested in the evidence of Russian crimes in form of geometrically precise point clouds showcasing the results of recent bombing; an urban analysts might query for low-resolution, mobile scans of a neighbourhood to reconstruct broader context. All in all, coherent metadata, enables managing digital artefacts at scale, catering to diverse use cases.

Ontologies and controlled vocabularies such as CIDOC CRM (Doerr, 2023), Getty Thesauri (Harpring, 2010), and IfcOWL (Beetz et al., 2009) are widely used to provide machine-readable semantic descriptions for the physical assets; while DOT ontology has been created to represent their damages (Hamdan et al., 2019). Existing metadata models already describe the provenance, acquisition and processing workflows of 3D digitisation (Homburg et al., 2021).

Building on these foundations, this research investigates how these developments can be combined and distilled into a lightweight profile that remains expressive while being practical for app-based, community-driven reality capture by non-expert contributors.

Thus, it bridges the gap between high-quality data requirements and low-threshold participative collection. The resulting metadata profile, as well as its ability to support representative retrieval scenarios are being evaluated through semi-structured expert reviews and by assessing its applicability within a workflow for community-based heritage recording.

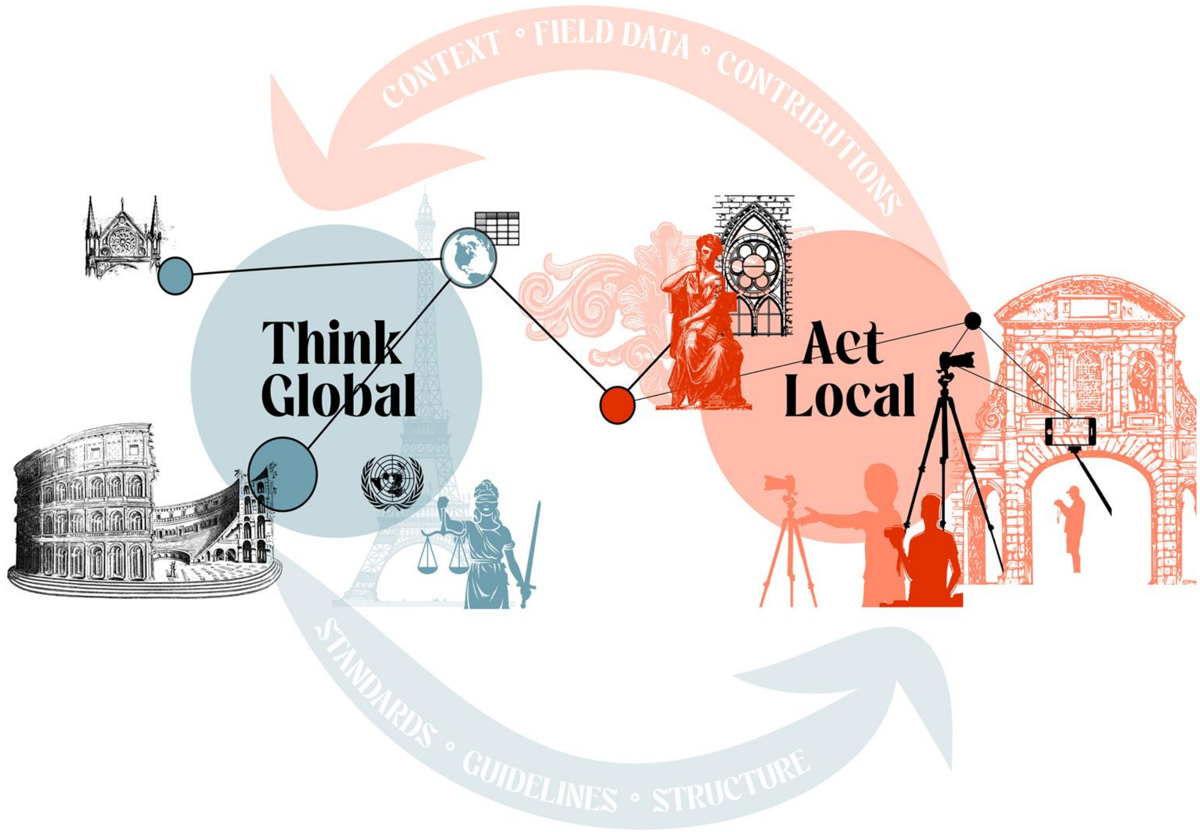

Fig. 1 A conceptual sketch of information exchange between the global heritage institutions and local enthusiasts.

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